Single-ended Chebyshev LPF Design

5-pole, 0.1 dB ripple, 15 MHz cutoff

Design Choices

Filter type: Chebyshev with 0.1 dB passband ripple. Chebyshev gives steeper rolloff than Butterworth at the cost of some passband ripple. 0.1 dB is small enough to be inaudible / invisible on instrumentation, while still giving meaningful rolloff improvement.

Order: 5 (5 reactive elements). Gives ~38 dB at 2×fc (28.4 MHz, 2nd harmonic) and ~58 dB at 2.84×fc (42.6 MHz, 3rd harmonic). Higher orders (7 or 9) increase rolloff but require more parts and tighten the tolerance requirements.

Topology: CLC (capacitor input/output). Shunt capacitors at both ends, series inductors in middle, third shunt capacitor in centre. Five elements: C1, L2, C3, L4, C5.

Cutoff: 15 MHz (about 5.6%% above the 14.2 MHz operating frequency). This gives good passband performance at 14.2 MHz with still-aggressive harmonic rejection above.

Component Values

Both systems use identical topology -- only component values scale with the chosen impedance. Inductor values scale as R0; capacitor values scale as 1/R0.

Comp	50 Ω system	300 Ω system	Nearest std (silver mica)
C1	243.4 pF	40.6 pF	240 pF / 39 pF
L2	727 nH	4.365 µH	wound on T-50-6 / T-68-2
C3	419.1 pF	69.9 pF	420 pF / 68 pF
L4	727 nH	4.365 µH	wound on T-50-6 / T-68-2
C5	243.4 pF	40.6 pF	240 pF / 39 pF

Schematic Topology

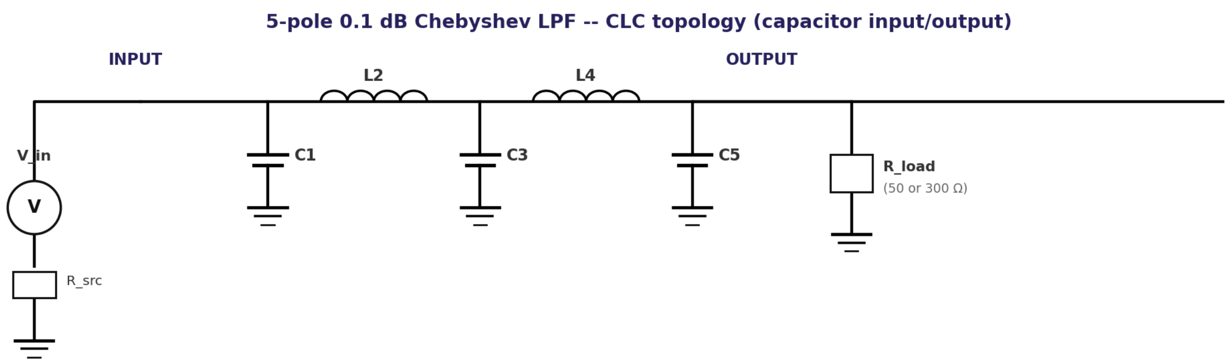

Figure 1 -- Single-ended Chebyshev LPF, CLC topology. Both 50 Ω and 300 Ω versions use this same topology with the component values from the table above.

Predicted Response

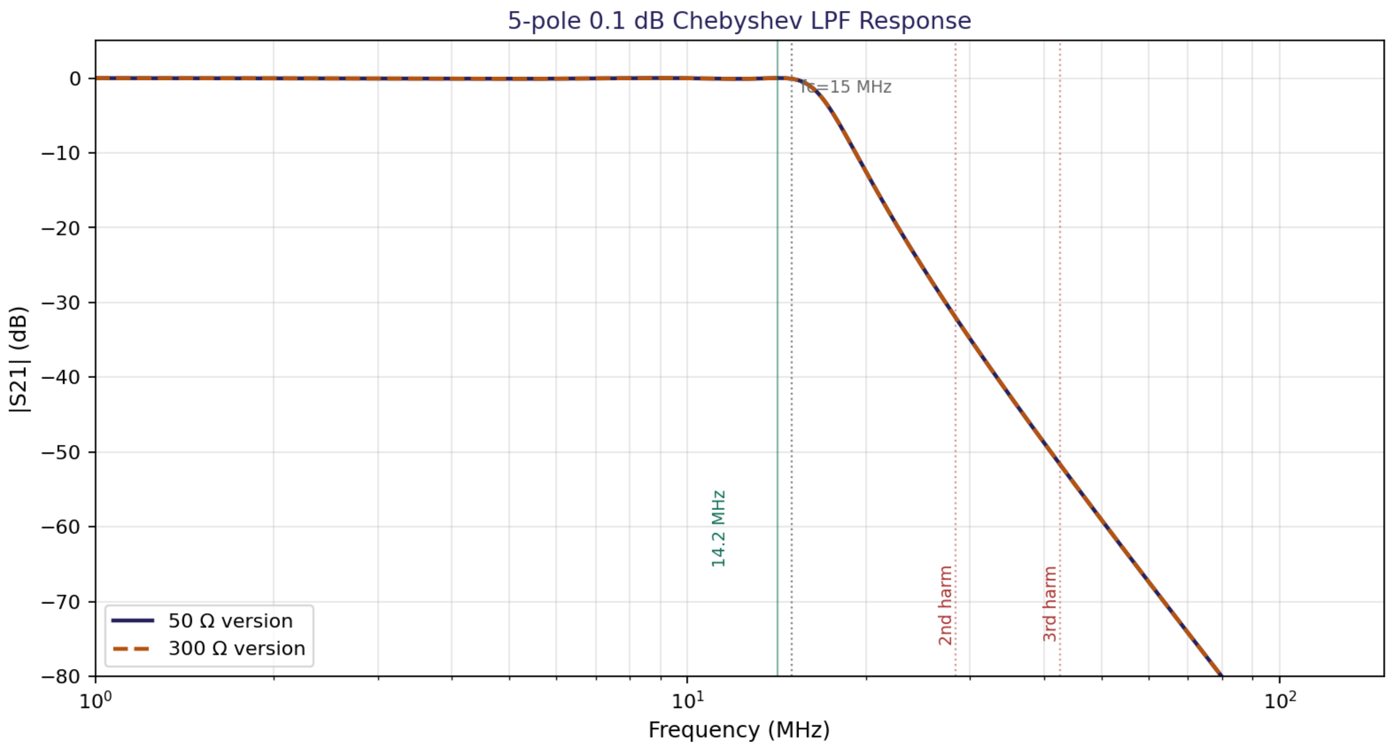

Figure 2 -- |S21| vs frequency for both impedances. The two responses are essentially identical (as they should be -- the filter shape is set by the ratios, not the absolute impedance). The 14.2 MHz operating frequency falls well in the passband; 2nd and 3rd harmonics fall in the stopband.

Tolerance Analysis

Real components have manufacturing tolerances. Effects on the filter response can be significant, especially near the cutoff where the response shape is most sensitive.

Assumptions used in the Monte Carlo analysis below:

* All capacitors: ±5%% uniform distribution (silver mica is typically ±5%%; NPO ceramic can be ±1%%)

* All inductors: ±5%% uniform distribution (hand-wound on iron powder cores, typical achievable accuracy)

* 200 runs per impedance, all 5 components varied independently

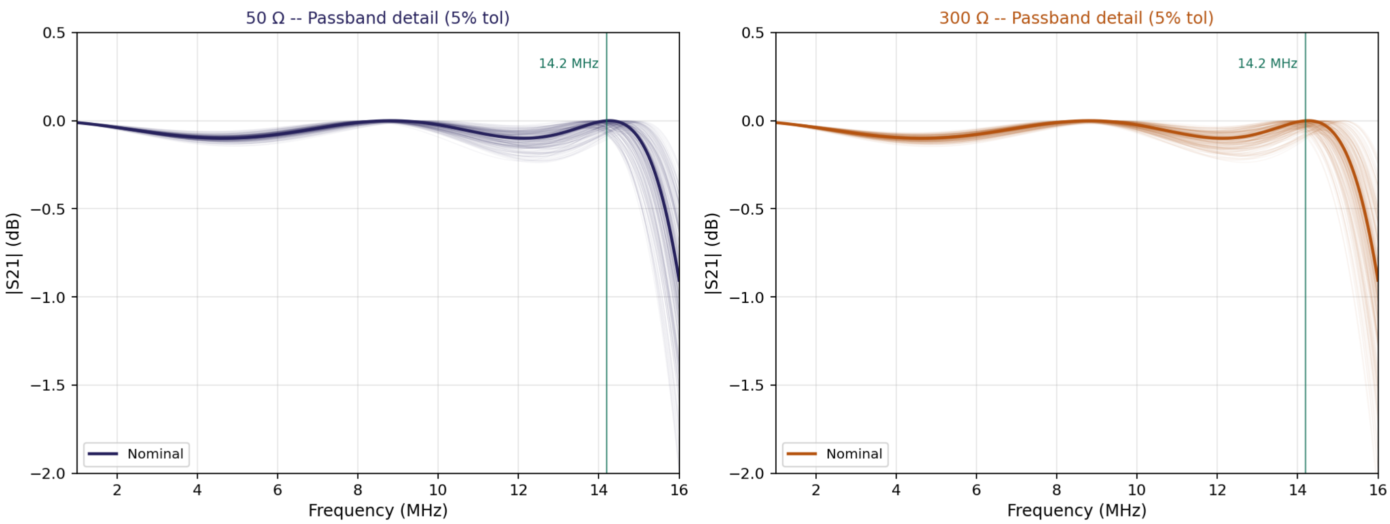

Figure 3 -- Passband detail showing ripple variation. The 0.1 dB nominal ripple becomes a broader band of possible responses under tolerance. At 14.2 MHz, expected insertion loss spread is ~±0.3 dB.

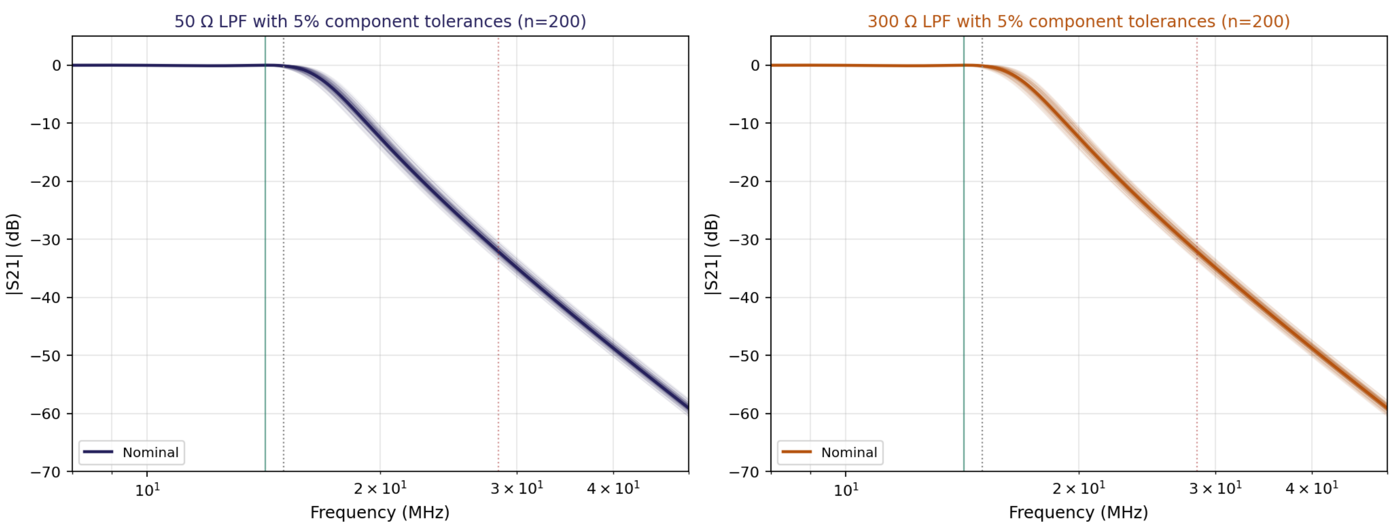

Figure 4 -- Stopband detail showing harmonic rejection spread. The nominal 38 dB at 28.4 MHz can vary by several dB worst-case but is reliable in the 30-45 dB range.

Key takeaway: with 5%% components the design is robust. Worst-case cutoff shift is ±500 kHz or so, which is well clear of the 14.2 MHz operating frequency. Stopband rejection variations of a few dB don't significantly change harmonic suppression performance.

50 Ω vs 300 Ω: Practical Tradeoffs

50 Ω version:

* Caps: 240 / 420 / 240 pF -- need higher value capacitors, but these are standard silver mica values

* Inductors: 728 / 728 nH -- very low values, may need careful air-wound or small toroid construction

* Compatible with coax (RG-58, RG-8) feedlines

* Needs 6:1 balun at PA output to convert 300 Ω balanced to 50 Ω unbalanced (4:1 if PA were 200 Ω, but 300:50 is 6:1 impedance, sqrt(6):1 = 2.45:1 turns ratio)

300 Ω version:

* Caps: 40 / 70 / 40 pF -- small values, easily achieved with silver mica or NPO ceramic, may need parallel combinations for accuracy

* Inductors: 4.37 / 4.37 µH -- comfortable values, easy to wind on T-50-2 or T-68-2 cores (~25 turns)

* Output goes directly to 300 Ω balanced twin-lead to dipole

* Simpler matching from PA (1:1 with the link winding ratio chosen for 300 Ω) -- no additional balun needed

Recommendation

For the design as originally conceived (push-pull PA → balanced twin-line → folded dipole), the **300 Ω version is the natural choice**. The component values are more practical (larger inductors, smaller capacitors) and no additional balun is needed at the LPF output.

The 50 Ω version is preferable if you want flexibility for coax feedlines or want to add an external 50 Ω antenna tuner. In that case the balun for converting from the PA's balanced output to 50 Ω unbalanced can be the link winding on the output tank itself.

> My suggestion: **build and simulate both as separate subcircuits**. They are small (5 parts each) and the tolerance analysis below is easy to repeat for either. Try both in simulation, decide based on whichever gives better predicted return loss in your specific PA loading scenario.

Simulating Tolerance in ngspice

For Monte Carlo analysis in ngspice via QUCS-S, use the .control block at the end of the netlist:

```
.control
* Monte Carlo loop, 100 runs
let mcruns = 100
let i = 0
dowhile i < mcruns
  alter @c1[capacitance] = {240e-12 * (1 + 0.05*(2*rnd-1))}
  alter @l2[inductance] = {730e-9 * (1 + 0.05*(2*rnd-1))}
  alter @c3[capacitance] = {420e-12 * (1 + 0.05*(2*rnd-1))}
  alter @l4[inductance] = {730e-9 * (1 + 0.05*(2*rnd-1))}
  alter @c5[capacitance] = {240e-12 * (1 + 0.05*(2*rnd-1))}
  run
  let i = i + 1
end
.endc
```

Use .AC analysis (not .TR) for filter response measurements. The AC analysis gives proper magnitude and phase across frequency. To extract just the S21 response, probe the load voltage with a known AC source amplitude.